

SCAI 2026 | TRACK 1 — SMART SENSING & INTELLIGENT ELECTRONIC SYSTEMS

DisasterShield

AI-Powered Multi-Hazard Early Warning System with Edge Intelligence

Connected Intelligence for Climate Resilience

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The Problem

Climate change is outpacing our ability to warn communities in time



Recurring hazards

Floods, wildfires, droughts across India



High economic loss

Crops, property & livelihoods at risk yearly



Too slow to act

Centralized systems can't give local alerts

*Existing early-warning systems are expensive, centralized, and cannot deliver the localized, real-time alerts that **vulnerable rural communities and farmers actually need.***

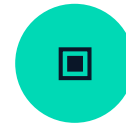
Proposed Solution

A low-cost, edge-AI system that watches, senses, and warns — in real time



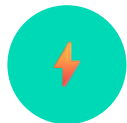
Real-Time Vision

Trained AI model detects fire & smoke from live camera feed



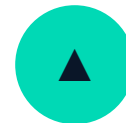
Sensor Fusion

Combines vision with temperature, humidity, rain & soil data



Edge-Ready

TensorFlow Lite model runs offline, no cloud dependency

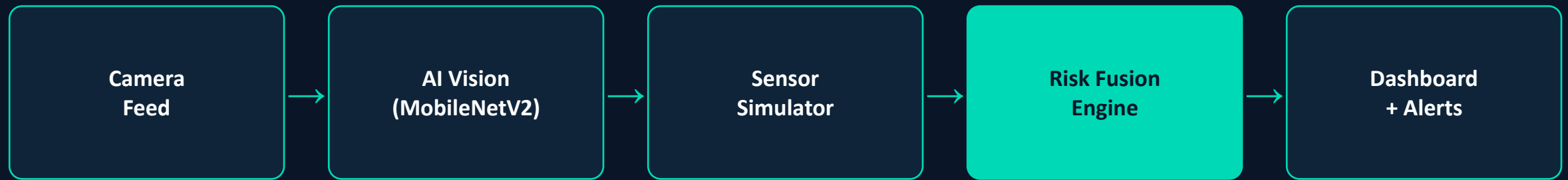


Multi-Hazard

Single risk engine covers fire, flood & drought signals

System Architecture

From live camera frame to actionable alert



Vision confidence and simulated sensor trends are combined by the Fusion Engine into one Low / Medium / High risk level, shown live on the dashboard.

AI & Computer Vision

MobileNetV2 transfer learning, trained on a real fire/non-fire dataset

99%

Training Accuracy

97.99%

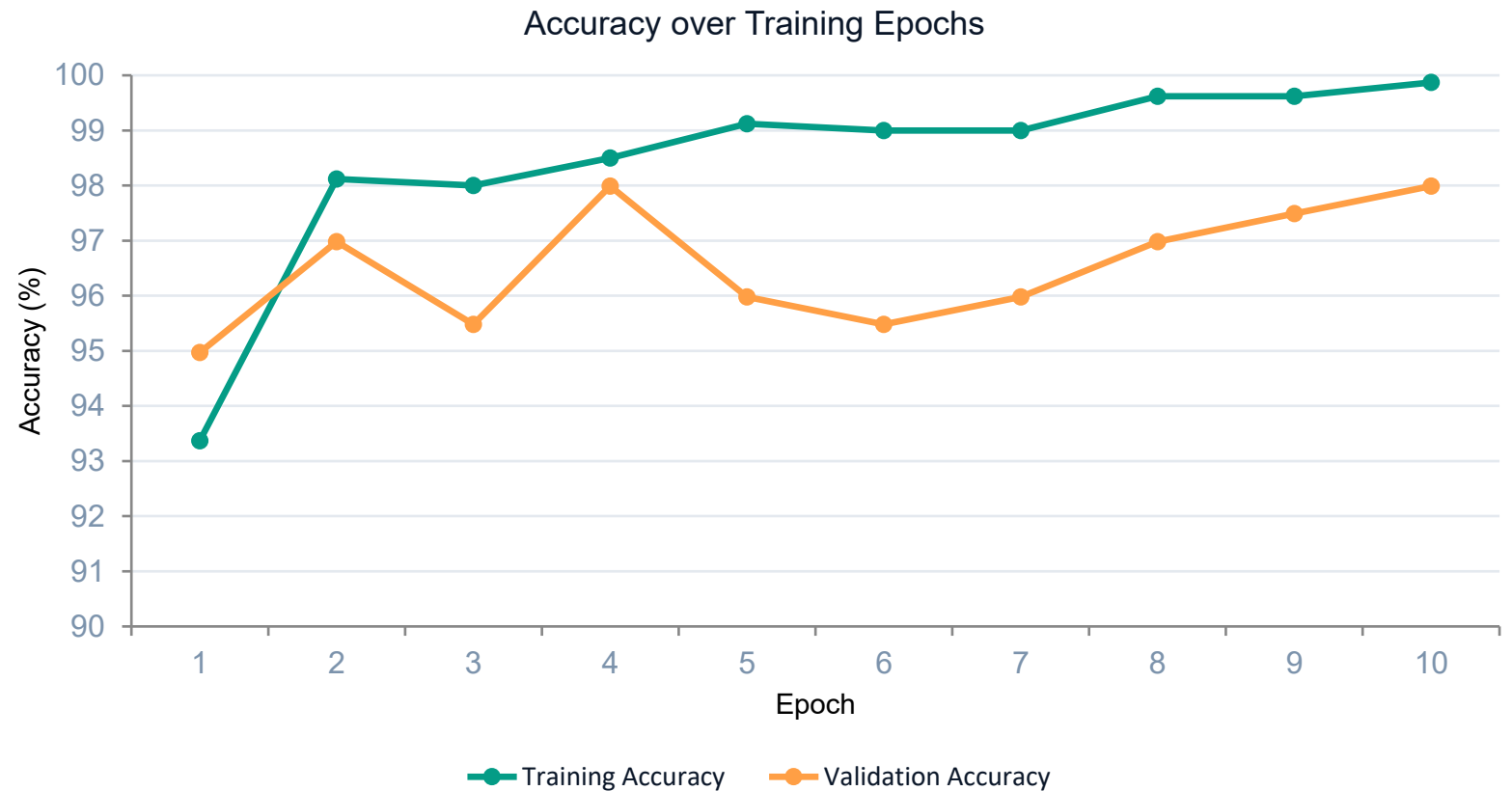
Validation Accuracy

10

Training Epochs

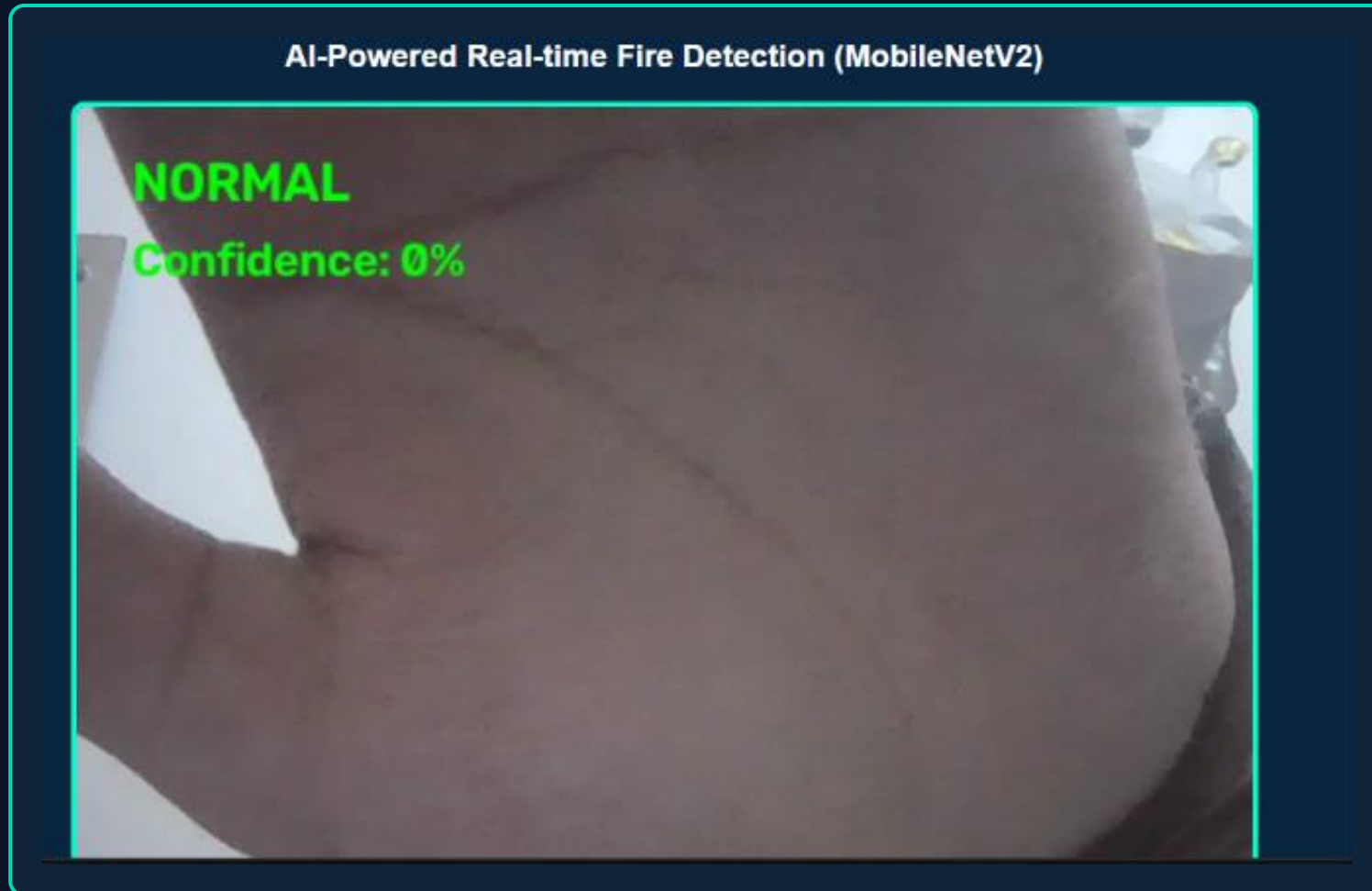
9.2 MB

TFLite Model Size



Live Demo

Watch the system detect fire/smoke and raise an alert in real time



1

Live camera feed

Streamed via Flask, processed frame-by-frame

2

AI confidence score

MobileNetV2 model output, updated every frame

3

Instant risk label

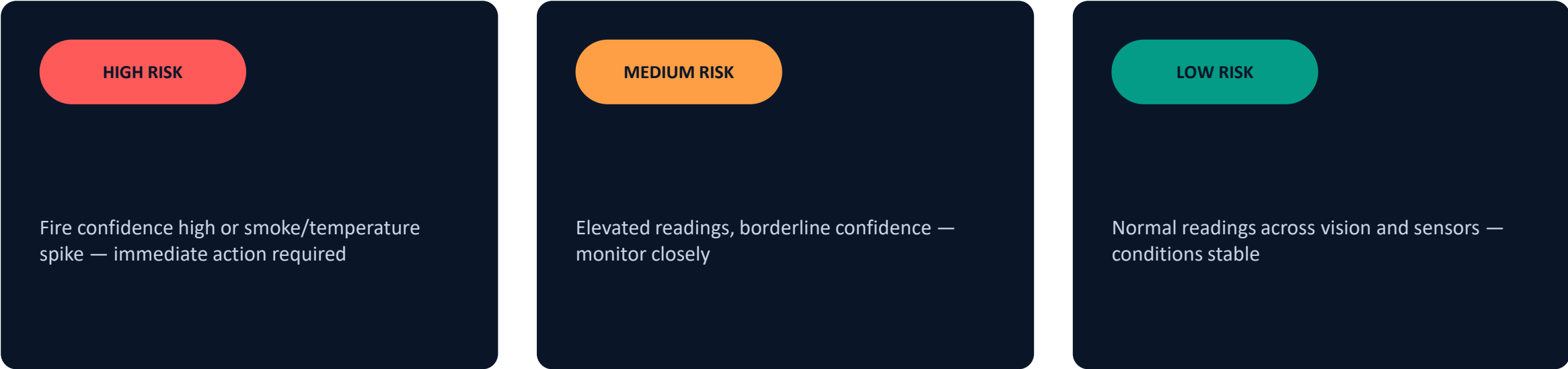
NORMAL / MEDIUM / HIGH RISK shown in real time

Multi-Hazard Risk Engine

Vision confidence + simulated sensors fused into one risk level



↓ *fused with AI vision confidence* ↓



Technologies & Implementation

Built entirely in Python, edge-deployment ready



Python 3.13

Core application language



OpenCV

Camera capture & frame processing



TensorFlow Lite

Edge-optimized AI inference



Flask

Live web dashboard



Open Source on GitHub

github.com/HarshParmar029/DisasterShield-SCAI2026

Edge-ready today — designed to run on Raspberry Pi 4/5 with the same TFLite model.

Current Achievements

What's built, tested, and working today — not just planned



Working live dashboard with real-time camera feed



Trained AI model — 97.99% validation accuracy



Real-time multi-hazard risk fusion engine



Complete GitHub repository with training notebook



Confusion matrix & sample predictions validated



TensorFlow Lite export — ready for edge deployment

Results & Impact

A working system with numbers to back it up

97.99%

Validation accuracy on held-out fire/non-fire images

< 1 sec

Detection latency per frame on a laptop CPU

9.2 MB

Compact TFLite model — fits on edge hardware

Societal Impact

Faster, localized alerts mean rural communities and farmers can protect lives, crops, and property before a hazard escalates — without expensive centralized infrastructure.

Future Scope

From prototype to field-deployable system

1

Real Hardware Sensors

Deploy on Raspberry Pi with DHT22, MQ-135 and soil sensors

2

Remote Sensing

Add NDVI/satellite-based wide-area risk mapping (Track 3 tie-in)

3

Connectivity

LoRaWAN for long-range, low-power alert transmission

4

Mobile App

Push notifications for field teams and affected communities

Connected Intelligence

for a Safer, More Resilient India

Thank you.

Questions & Discussion

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